

Wage Determination and Returns to Skill

Labor I — Week 5

Teaching deck draft

Central question and motivation

- Week 5 asks how markets translate accumulated skill into observed wages.
- A schooling premium can bundle productivity, selection, sorting, amenities, and rents.
- The recurring question is not just “is the effect causal?” but “which return object are we estimating?”

Week 4 to Week 5 bridge

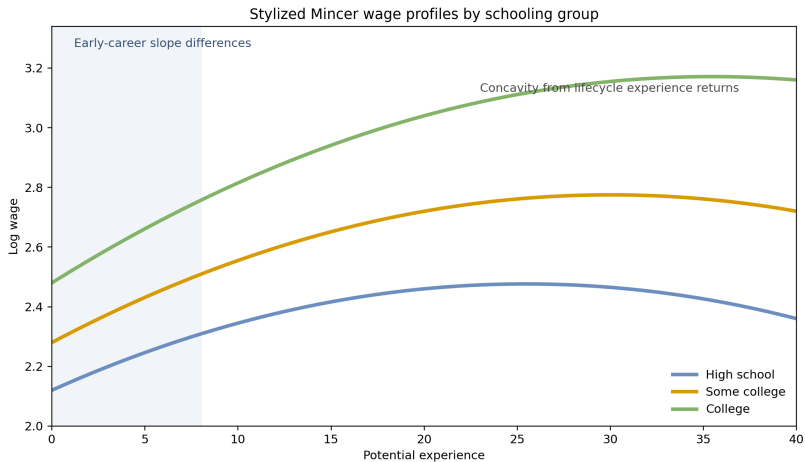
- Week 4 treated skill as an accumulated state variable.
- Week 5 treats wages as equilibrium outcomes that price that state only imperfectly.
- We therefore move from human-capital accumulation to wage measurement, identification, and sorting.

Mincer benchmark

$$\log w_i = \alpha + \rho s_i + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i$$

- Useful benchmark decomposition of wage differences by schooling and experience.
- ρ is descriptive unless stronger assumptions make it causal.
- Wage returns are not the same as productivity returns, especially once hours and job assignment matter.

Stylized Mincer profiles



Selection and heterogeneous gains

$$Y_i(s) = \mu_i + g_i(s)$$

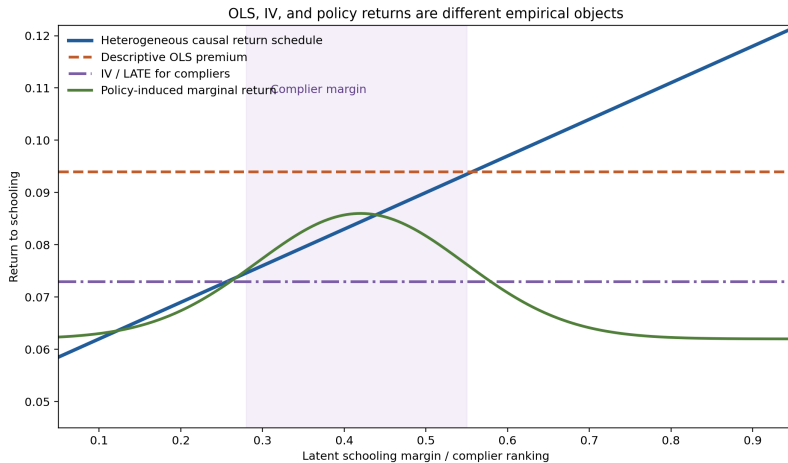
- Selection is not only ability bias in levels.
- Comparative advantage means workers sort on gains as well as on baseline productivity.
- OLS, IV, and structural approaches answer different questions once $g_i(s)$ varies across people.

What does IV estimate?

$$\beta^{IV} = E \left[\frac{Y_i(1) - Y_i(0)}{S_i(1) - S_i(0)} \middle| S_i(1) > S_i(0) \right]$$

- Compulsory-schooling IV identifies a local return for the margin shifted by the policy.
- Oreopoulos is useful because the first stage is economically meaningful.
- Stephens and Yang remind us that familiar instruments can still be trend-sensitive.

OLS versus IV / LATE



Parameter map for returns-to-schooling research

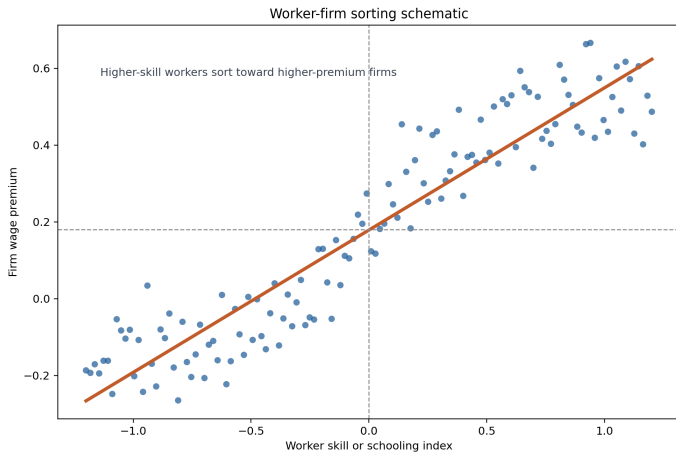
Object	Design	What it measures	Main threat
OLS premium	Mincer regression	Conditional schooling-wage association	Selection and sorting
ATE	Strong structure / experiment	Mean causal gain	External validity
LATE	Compulsory-schooling IV	Return for compliers	Margin-specific interpretation
Marginal policy effect	LIV / structural	Policy-induced local gain	Support and functional form
Firm-mediated return	Matched worker-firm data	Share of premium via better firms	Firm effects bundle rents and selection

Wages are also shaped by firms and places

$$\log w_{it} = \alpha_i + \psi_{j(i,t)} + x'_{it}\beta + \varepsilon_{it}$$

- Wage dispersion can reflect worker effects, firm premia, and market assignment.
- Education may raise wages partly by improving access to higher-paying firms or cities.
- This is why Week 5 is also the bridge into inequality and mobility work.

Worker-firm sorting schematic



Design buckets and interpretation

- Descriptive wage equations: good for measurement, weak for causal interpretation.
- Compulsory-schooling IV: local return for the induced schooling margin.
- Matched worker-firm decompositions: map wage dispersion across workers and firms.
- Spatial sorting designs: show that some skill returns are place-mediated.

Bridge to Week 6 and Week 8

- Week 6: households change schooling choices, labor supply, and observed wage offers.
- Week 7: amenities imply that wages are incomplete welfare objects.
- Week 8: inequality work needs worker, firm, and market decompositions rather than one residual “skill premium.”

- Mincer is a benchmark, not the truth.
- Changing the design changes the estimand.
- Returns to schooling can operate through firms and places, not only within-worker productivity.
- Wage determination is the discipline that connects human capital to inequality.

Appendix: optional extension block

- Worker-firm decompositions in the spirit of AKM and modern firm-inequality work.
- Spatial sorting and diverging city returns to skill.
- Optional lab reflection: how much of a schooling premium might be mediated by better firms or places?